THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL 1 EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

ENGINEERING MECHANICS AND MATERIALS – CIVL1013

DATE: 15 May, 2012 TIME: 9:30 am – 12:30 pm (3 hours)

Answer any 2 questions from Section A and any 2 questions from Section B.
All questions carry equal marks.
Use a separate answer book for Section A and Section B.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

You are required to answer 2 out of 3 questions in Section A.

Question 1

Consider the simply-supported beam with overhang shown in Figure 1 which is subjected to the following loading:

- Uniformly distributed load (intensity w) applied along whole length of beam,
- Point load (magnitude P) applied at free-end of overhang, and
- Couple (magnitude M_1) applied at point B.

(a) Draw the shear force and bending moment diagrams when only the uniformly distributed load (w) is applied to the beam. (**8 marks**)

(b) Draw the shear force and bending moment diagrams when only the point load (P) is applied to the beam. (**6 marks**)

(c) Draw the shear force and bending moment diagrams when only the couple (M_1) is applied to the beam. (**6 marks**)

(d) Utilising the Principal of Superposition, draw the shear force and bending moment diagrams when all loads (w, P and M_1) are applied to the beam simultaneously. All possible solutions need to be shown. (**5 marks**)

Note:
- all bending moment diagrams are to be drawn on tension side
- position and magnitude of maximum bending moment to be shown on all bending moment diagrams

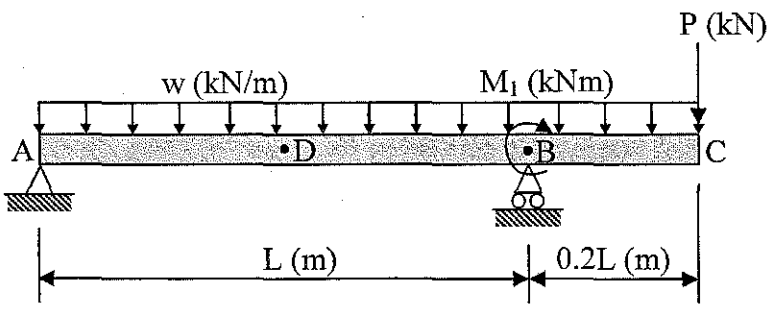

Figure 1.

Question 2
Consider the simply-supported beam with overhang shown in Figure 1 which is subjected to loads w, P and M_1.

(a) Determine all reactions (include suitable check). **(6 marks)**

(b) Derive an expression for the deflection curve along the entire length of the beam (utilise Macaulay's method). **(14 marks)**

(c) If L = 10 m, w = 8 kN/m and M_1 = 40 kNm, determine the magnitude of the vertical force P to produce zero deflection at the mid-span of beam A-B (i.e. point D in Figure 1).
(5 marks)

Question 3
Consider the following transformation equations for plane stress.

$$\sigma_{x_1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\sigma_{y_1} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2}\cos 2\theta - \tau_{xy}\sin 2\theta$$

$$\tau_{x_1 y_1} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

(a) Derive equations for calculation of the maximum and minimum principal stresses.
(13 marks)

(b) Prove that the maximum shear stress planes occur at 45° to the principal stress planes.
(12 marks)

- END OF SECTION A -